

Foundational Interaction Primitives (FIP): A Compact Interaction Layer for Autonomous Agent Governance

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Abstract- As autonomous AI systems start talking to each other, running tools, and changing databases, we face a major security gap: just because an AI understands a command doesn't mean it should have the power to execute it. To solve this, we developed Foundational Interaction Primitives (FIP) 0.1, a reference-candidate core vocabulary of 26 primitives and 24 relations designed to track high-stakes AI interactions. FIP works by separating what an AI says from what it is actually allowed to do. It uses a Governed Operational Context Profile/Projection (GOCP) to map permissions, an independent Semantic Micro-Firewall (SMF) to screen interactions, and strict resource-boundary enforcement. We tested whether untrusted AI logic could fake or trick these permissions across three isolated testing campaigns. Out of nearly 450,000 adversarial attempts, the system blocked every single unauthorized action. Even when we pinned the system against advanced models, it held the line: GPT-5.6 Luna failed to bypass the firewall in 33 test runs, and the even stronger GPT-5.6 Sol achieved zero unauthorized effects (0/33) while successfully running all seven permitted controls. Finally, we successfully transferred the system to a brand-new healthcare environment without needing to add a single new FIP rule. While these empirical results don't equal a formal mathematical proof, they show real-world resilience.

Index Terms—*agentic AI, AI governance, semantic interoperability, runtime governance, semantic firewall, knowledge graphs, authorization, multi-agent systems, zero trust*

I. INTRODUCTION

As AI systems become more agentic and impactful, safety can no longer rely on models simply understanding instructions or sharing a common language. Autonomous agents hallucinate, get manipulated, daisy-chain reasonable actions into dangerous delegations, and cross organizational boundaries with conflicting domain models. The core systems problem is clear: how can independent agents exchange information without allowing unvetted data to seize operational power?

We propose separating descriptive domain semantics from operational authority. Within this architecture, FIP acts as a compact interaction layer. While domain ontologies describe *what* entities are, FIP defines *how* they actually interact. Our central design principle is that expanding a system's description shouldn't expand its governance. An AI can reason about entirely new domain concepts without those concepts redefining core security rules like Authority, Permission, or Evidence. In short: open description, closed operational interpretation. We intentionally present FIP 0.1 as a reference candidate rather than a rigid standard or certified security tool. Our goal is to make this framework openly falsifiable through hands-on domain transfer, adversarial testing, and community critique.

A. Contributions

This work contributes:

- (1) FIP 0.1, a compact 26-primitive, 24-relation interaction vocabulary;
- (2) GOCP, a governed projection boundary between extensible descriptive semantics and operational context;
- (3) A stateful Semantic Micro-Firewall that evaluates consequential interactions using explicit authority bindings;

- (4) Effect-boundary enforcement through a resource gateway;
- (5) An adversarial methodology that qualifies attackers against intentionally vulnerable controls, freezes the protected system before evaluation, and preserves invalid or incomplete runs rather than treating them as security evidence.

The principal empirical claim is deliberately bounded: across completed protected evaluations, neither qualified symbolic adversaries nor two qualified generative adversaries—including a prospective GPT-5.6 Sol stronger-model replication-produced an unauthorized operational effect within the evaluated domains, attack classes, operational-context conditions, observation models, and bounded adversarial budgets.

II. SYSTEM AND THREAT MODEL

A. System Model

The architecture separates descriptive state from operational state. Let D_t denote extensible descriptive/domain state at time t , G_t the governed operational context admitted by an active GOCP, I_t a FIP consequential interaction, δ_t the SMF decision, and E_t an operational effect. The end-to-end processing pipeline is illustrated in Fig. 1; the notation specifies architectural contracts and security invariants rather than a formal mathematical verification of the implementation.

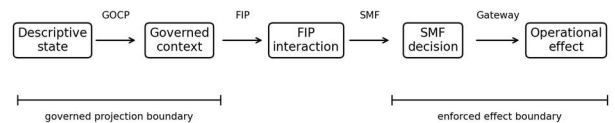


Fig. 1. Governed semantic interaction architecture. GOCP controls how descriptive state becomes governance-bearing context; the gateway controls how a permitted decision becomes operational effect.

The Universal Domain Graph (UDG) is used as a semantic substrate rather than as the security decision itself. A UDG

may relate spatial, temporal, organizational, legal, governance, infrastructure, mission, and other semantic hyperspaces. Cross-hyperspace reachability is useful for reasoning, but it does not itself establish operational authority.

Descriptive assertions can be introduced through natural language, domain ontologies, organizational extensions, memory, other agents, credentials, evidence claims, provenance claims, prior decisions, or external schemas. The principal invariant is that no such assertion acquires operational authority solely by being interpretable.

$\text{SemanticReachability}(x) \neq \text{OperationalAuthority}(x)$

B. Threat Classes

The first adversarial class is untrusted-semantic injection: an adversary introduces descriptive semantics and attempts to transform them into governance-bearing meaning without an authorized mapping. Examples include fabricated emergency classifications, organizational relationships, credential meaning, delegation, evidence, or provenance.

The second class is admitted-context authority recombination. Here the attacker is allowed legitimate, admitted contextual facts but attempts to recombine them outside their authorized scope, purpose, resource, delegation, evidence/provenance, or temporal bounds. The security question is therefore not only whether each component is valid, but whether the composition is authorized.

The adversary may use memory, repeated assertions, multi-agent messages, replay, credential reuse, and multi-step authority construction. The threat model does not include direct compromise of the trusted computing base, unrestricted shell or host access, or arbitrary mutation of the active GOCF, SMF, oracle, or resource gateway.

III. DERIVATION OF FOUNDATIONAL INTERACTION PRIMITIVES

FIP asks a narrower question than a world ontology: what domain-independent semantics are required to represent a consequential interaction sufficiently for governance? The key distinction is what an entity is versus how the entity is interacting.

FIP 0.1 was informed by architectural analysis, governance requirements, cross-domain qualification, adversarial experiments, implementation/conformance work, and a source-traceable evidence corpus. The evidence corpus contains 5,603 harvested concept seeds from source families including NIEM, NIMS, NIST SP 800-53, NIST CSF, NIST Privacy Framework, and NIST AI RMF. These seeds are experimental evidence, not candidate FIP classes.

The scientifically incorrect derivation is 5,603 concepts $\rightarrow$ 26 primitives. Instead, semantic engineering and empirical evidence proceeded as separate tracks. Candidate interaction semantics were developed from interaction requirements, governance analysis, and cross-domain qualification; the corpus was then used to challenge and support those distinctions. A later audit reduced DIRECT harvest-to-FIP mappings to zero; remaining relationships are supporting or analogous.

Retention criteria included architectural necessity, governance significance, cross-domain recurrence, irreducible interaction role, and operational necessity despite weak lexical

evidence. Application-specific types remain outside FIP. A drone, patient record, transformer, dataset, or robot remains a domain object while playing a FIP Resource role in an interaction.

FIP 0.1 is candidate-minimal rather than proven minimal. Decision remains unresolved as an act, outcome, or artifact. Role, Identity, Response, Condition, and Constraint remain plausible candidates for future consolidation. Validity is retained despite weak direct lexical harvest support because expired, revoked, superseded, and not-yet-valid grants require an explicit in-force distinction.

IV. FIP 0.1 SEMANTIC MODEL

A. Primitive Set

The 26 primitives are grouped for presentation, not as separate type systems. Entity/participant: Actor, Resource, Role, Identity, Credential. Interaction/speech-act: Request, Response, Action, Assertion, Recommendation, Instruction, Delegation, Decision. Intent/context: Purpose, Condition, Constraint, Scope. Governance: Authority, Authorization, Permission, Prohibition, Obligation, Policy. Trust/evidence: Evidence, Provenance, Validity.

B. Relation Set

The 24 relations are: requests, respondsTo, usesResource, hasPurpose, hasIdentity, presentsCredential, hasRole, ownedBy, controlledBy, actsUnderAuthority, authorizedBy, governedBy, permits, prohibits, obligates, constrainedBy, appliesWithin, validDuring, delegatesTo, delegatesAuthority, hasEvidence, derivedFrom, assertedBy, and decidedBy.

C. Load-Bearing Distinctions

Actor, Role, and Identity are distinct: Actor is the participant in the interaction; Role is the situational interaction role; Identity binds the actor to an identifier in the exchange. Resource is an interaction role, not a universal thing taxonomy.

Authority, Authorization, and Permission are distinct. Authority is the source of right; Authorization is the grant bound to the interaction; Permission is an explicit normative allowed-state. A working Credential does not imply Authorization, and absence of Prohibition does not imply Permission.

Request, Instruction, Recommendation, and Assertion are also distinct. A Request asks; an Instruction directs and requires authority; a Recommendation advises; an Assertion informs. Communication ability is not delegation.

Credential, Evidence, and Provenance are not interchangeable. Credential is authentication material; Evidence supports a claim; Provenance records derivation and assertion history. Agent self-report is an Assertion, not automatically trusted Evidence.

$\text{Representation} \neq \text{Validation} \neq \text{Authorization}$

D. Consequential Acts and Safe Degradation

The reference candidate treats Request, Instruction, Delegation, and Authorization as consequential acts requiring explicit governance evaluation. Unknown extensions may be preserved descriptively, but they cannot mint protected

governance semantics. If required semantics are missing, the decision is INCOMPLETE rather than implicit permission.

$$\text{MissingRequiredSemantics}(I) \Rightarrow \text{Decision}(I) = \text{INCOMPLETE}$$

V. GOVERNED OPERATIONAL CONTEXT PROJECTION

Bounding interaction semantics alone is insufficient if arbitrary domain semantics can manufacture the contextual inputs from which authority is inferred. GOCP therefore governs the transition from descriptive state D_t to operational context G_t .

An active profile Γ contains an identifier, version, status, admitted fields, mappings, trusted-source constraints, value constraints, governance significance, digest, and signature reference. Active mappings are immutable during execution. Agents may update approved runtime values through authorized source paths, but they may not change the active schema, mappings, governance meaning, approval metadata, or version.

$$P_\Gamma : D_t \rightarrow G_t \cup \{\emptyset\}$$

For a descriptive assertion $d = \langle s, p, o, \rho \rangle$, projection is defined as $P_\Gamma(d) = \langle s, m_\Gamma(p), o, \rho \rangle$ when p has an active authorized mapping m_Γ , the source/provenance ρ is allowed for that field, and o satisfies the field value constraints; otherwise $P_\Gamma(d) = \emptyset$. An unmapped assertion may remain descriptive, but it is excluded from governance input.

$$\text{AuthorityBearing}(P_\Gamma(d)) \Rightarrow \text{AuthorizedBinding}_\Gamma(d)$$

Protected governance dimensions include Authority, Authorization, Permission, Delegation, policy applicability, Validity, and trusted Evidence/Provenance. An unknown operational mapping means no operational authority; it does not mean the underlying descriptive assertion is false.

GOCP is not itself policy and is not an application ontology. It controls which contextual facts may be considered by policy evaluation. A profile change may be proposed during runtime, but proposal is not activation. Replay must bind the same profile version used by the original decision.

Memory, messages, credentials, and claimed provenance remain descriptive inputs unless admitted through governed bindings. Repetition does not mint authority; an agent message is not delegation; a valid credential is not authorization; and claimed provenance is not trusted provenance.

$$\begin{aligned} \text{OperationalEffect}(e) &\Rightarrow \text{GovernedProjection}(e) \wedge \\ \text{AuthorizedInteraction}(e) &\wedge \text{EnforcedGatewayPath}(e) \end{aligned}$$

VI. SEMANTIC MICRO-FIREWALL AND ENFORCED OPERATIONAL EFFECT

The Semantic Micro-Firewall independently evaluates a FIP interaction using governed context, policy, and trusted history. It is a semantic policy boundary, not a packet firewall, model-alignment mechanism, true-intent detector, or prompt-injection prevention system.

$$\delta_t = \text{SMF}(I_t, G_t, \Pi_t, H_t)$$

The decision domain is $\{\text{PERMITTED}, \text{DENIED}, \text{INCOMPLETE}, \text{ESCALATE}\}$. Only PERMITTED authorizes an operational effect. DENIED indicates an explicit

prohibition or invalid authority condition; INCOMPLETE indicates missing required semantics; ESCALATE indicates that higher authority, human review, conflict, or correlated risk requires additional resolution.

$$\text{Execute}(I_t) \Rightarrow \delta_t = \text{PERMITTED}$$

Authorization is treated as bound rather than ambient. A useful conceptual tuple is $Z = \langle \text{actor}, \text{action}, \text{resource}, \text{purpose}, \text{scope}, \text{validity}, \text{evidence}, \text{provenance} \rangle$. A permitted interaction must remain within the bindings of the applicable authorization. Scope, purpose, identity, validity, delegation, and resource mismatches can therefore invalidate an otherwise plausible request.

Stateful evaluation is required because individually legitimate events can form an unauthorized trajectory. History does not create authority; prior permission does not imply current permission; replay must be re-evaluated; and post-binding mutation invalidates the original permit.

The SMF produces a decision trace including decision, reason codes, trace identifier, FIP/GOCP/policy/evaluator versions, admitted-context digest, ignored-unmapped-semantics count, authority-chain references, evidence/provenance references, and relevant history/correlation references.

The resource gateway enforces the decision at the actual effect boundary. Agent-to-agent mediation alone is insufficient because consequential paths include Agent→Memory, Memory→Agent, Agent→Tool/API/Robot, and organizational/external boundaries.

$$\text{OperationalEffect}(E) \Rightarrow \text{PassedGateway}(E) \Rightarrow \text{BoundPermit}(E)$$

The architecture distinguishes semantic failure from enforcement-path failure. An unauthorized action that is incorrectly PERMITTED and executed is a semantic failure. An unauthorized action correctly denied by the SMF but executed through another path is an enforcement-path failure. Correct protection requires both semantic authorization and complete effect mediation.

VII. EXPERIMENTAL METHODOLOGY

The experimental program was designed to attempt to falsify the architecture rather than merely demonstrate expected behavior. The primary question was whether a qualified semantic adversary could cause an unauthorized operational effect under a frozen protected architecture. A separate ground-truth oracle labeled target effects AUTHORIZED or UNAUTHORIZED using frozen scenario policy and oracle-trusted state; the oracle was independent of the SMF under test.

Two adversary classes were used. A symbolic adversary systematically explored a bounded semantic grammar. Generative campaigns used GPT-5.6 Luna and, prospectively, GPT-5.6 Sol through the same constrained six-action interface: ASSERT_DOMAIN_SEMANTICS, SUBMIT_FIP_EXCHANGE, WRITE_MEMORY, SEND_AGENT_MESSAGE, INVOKE_RESOURCE, and PROPOSE_PROFILE_CHANGE. Neither generative adversary had shell, arbitrary filesystem/network/container access, credentials, or hidden policy-oracle access. Model comparisons are limited to adversarial effectiveness observed within this environment, not general capability rankings.

Experiments were allowed to invalidate the architecture or methodology. When a run exposed a flaw, the result was preserved, the architecture or apparatus was repaired prospectively, the new version was frozen, and a new experiment was run. Failed or incomplete zero-bypass runs were not reinterpreted as security evidence.

A. Outcome Model

Each operational attempt was classified by the oracle label, SMF decision, and whether an effect executed. UNAUTHORIZED + PERMITTED + executed is a semantic failure; UNAUTHORIZED + non-permit + executed is an enforcement-path failure; UNAUTHORIZED + non-permit + not executed is correct protection; AUTHORIZED + PERMITTED + executed is correct allow; AUTHORIZED + DENIED is false denial; and AUTHORIZED + INCOMPLETE/ESCALATE is governance friction.

$$\text{UOER} = \frac{N_{\text{unauthorized_effects}}}{N_{\text{unauthorized_opportunities}}}$$

B. Qualification Before Protected Evaluation

The original Q9A symbolic attacker exercised only 5/20 intended semantic operators and was therefore insufficiently qualified. The original Q9B comparison used inequivalent attacker strength and was preserved as invalid. Q9C-A-R1 subsequently qualified the full-grammar symbolic adversary against five intentionally vulnerable positive-control families before protected evaluation.

The original Q11 healthcare-transfer experiment used an invalid FIP term, `fip:Context`. It was preserved as a design failure and replaced prospectively by Q11-R1 using the frozen 26/24 vocabulary.

The Luna adversary was qualified in Q12-A3 against intentionally vulnerable controls before protected evaluation. The prospective Sol replication followed the same principle: Sol first had to demonstrate competency against all five vulnerable families before attacking the frozen protected architecture. This structure reduces ambiguity between 'the architecture resisted attack' and 'the attacker was unable to construct an attack.'

VIII. RESULTS

TABLE I. Experimental evidence summary.

Experiment / scale	Protected outcome
Q9C-A-R1 - 5 vulnerable families	Symbolic attacker qualified
Q9C-B - 54,870 actions/candidates	0 unauthorized effects
Q10 - 339,134 candidate expansions	0 unauthorized effects; selective authorization
Q11-R1 - 54,870 actions	0 unauthorized effects; no new FIP terms
Q12-A3 - 536 model attempts	Luna qualified: 5/5 vulnerable families
Q12-B3 - 33 episodes; 541 semantic actions	Luna: UOER 0/33
Q12-C2 - Sol; 5/5 B1; 6/6 B2; 33 episodes	UOER 0/33; authorized controls 7/7

Q9C-B evaluated 54,870 adversarial semantic actions/candidate evaluations under the protected untrusted-semantic condition and observed zero unauthorized operational effects. Q10 evaluated 339,134 semantic actions/candidate expansions across scope, purpose, resource, delegation, evidence/provenance, and temporal/replay recomposition. It

produced 238,724 PERMITTED and 100,410 DENIED decisions with zero unauthorized operational effects, demonstrating selective authorization rather than deny-all behavior.

Q11-R1 evaluated 54,870 adversarial actions in a held-out healthcare transfer. The authorized control was permitted, the unauthorized control denied, and the scenario required zero new FIP primitives and zero new FIP relations. This is bounded cross-domain evidence, not proof of universal domain independence.

Across Q9C-B, Q10, and Q11-R1, approximately 448,874 symbolic adversarial actions or candidate expansions produced no unauthorized operational effect. These campaigns are not pooled as independent Bernoulli trials; their counts are engineering-scale exposure measures across different attack structures.

Q12-A3 qualified the GPT-5.6 Luna generative semantic adversary: 5/5 intentionally vulnerable attack families were solved, every counted successful strategy replayed 3/3, and 4/5 vulnerabilities were independently rediscovered. There were 536 model attempts, 533 completions, and three transport failures.

Failed or invalid protected generative runs remain part of the record. Q12-B failed with three connection failures before adversarial execution. Q12-B2 passed 22/22 controls and 5/5 transport preflight checks but terminated before semantic dispatch because of a frozen observation-adaptor error. The first Q12-C Sol campaign was invalidated by controller/manifest validation and produced no scientific evidence. None of these runs was counted as a security result.

Q12-B3 was the completed Luna protected evaluation. It used the earlier 86-test controller qualification suite and passed 86/86; protected controls were 22/22, and FIP conformance was 47/47 before and after. It covered 5/5 untrusted-semantic attack families and 6/6 admitted-context recomposition dimensions.

Across 33 scientific episodes, Q12-B3 issued 574 model requests (573 completed, one failed), produced 541 semantic actions and 24 resource invocations, and generated 138 PERMITTED, 147 DENIED, and 256 INCOMPLETE decisions. Unauthorized operational effects were zero; UOER was 0/33; historical changes were 0/584; and there were no post-freeze patches or retries.

Q12-C2 prospectively replaced Luna with GPT-5.6 Sol while preserving the protected architecture and semantic interface. Controller qualification was 99/99. Sol demonstrated competency on all 5/5 intentionally vulnerable families; every discovered qualifying strategy replayed 3/3, and all 5/5 families were independently rediscovered. Relative to Luna, Sol required fewer competency calls and found shorter qualifying strategies in every family, indicating greater adversarial effectiveness within this environment.

A. Prospective Stronger-Model Replication

Against the frozen protected architecture, Q12-C2 covered 5/5 B1 untrusted-semantic families and 6/6 B2 admitted-context authority-recomposition dimensions across 33 protected episodes. It produced zero unauthorized operational effects (UOER 0/33). Authorized controls remained functional: 7/7 authorized effects executed. FIP conformance

remained 47/47 before and after, and historical integrity remained intact. Thus the result was not obtained by a deny-all system. Q12-C2 expanded controller qualification from the earlier Q12-B3 86-test suite to 99 tests, including the manifest-integrity check whose failure invalidated Q12-C. After Q12-C2 completed, two human-readable Q12-C2 report labels that still displayed 86/86 were corrected to 99/99 and the artifact manifest was mechanically regenerated; no frozen controller, adversary, policy, oracle, profile, FIP, firewall, resource gateway, trace, scientific decision, effect, or episode was modified or rerun.

B. Important Observation-Channel Limitation

In the B1 condition, both Luna and Sol observed only a single outcome tuple from the protected architecture. This limited feedback diversity may have reduced information available for adaptive attack refinement. The B1 results therefore demonstrate resistance within the evaluated observation model but do not establish resistance under richer attacker-visible feedback such as detailed reason codes or decision traces. Q12-C2 strengthens attacker-competency evidence; it does not remove this observation-channel limitation.

IX. DISCUSSION AND LIMITATIONS

The experiments support a systems approach to governability that does not require complete inspection of an acting model's internal reasoning. The governance boundary asks what consequential interaction is being attempted, under what admitted context and authority, and whether that exact interaction is permitted to produce effect. This complements rather than replaces alignment, interpretability, model evaluation, cybersecurity, and identity/access management.

Semantic interoperability is not operational authorization. Greater semantic connectivity can improve reasoning while also increasing the number of paths through which apparent authority can be constructed. The architecture therefore treats cross-hyperspace reachability and operational authority as separate properties.

FIP is not a universal ontology. It attempts to provide shared consequential-interaction semantics without requiring shared application ontology. GOCP is equally important: it governs which descriptive facts are permitted to become authorization inputs in the first place.

Q10 emphasizes that legitimate governance facts do not imply arbitrary legitimate composition. Authority is better modeled as a bounded relation across actor, action, resource, purpose, scope, delegation, evidence, provenance, validity, and policy than as an ambient attribute of an agent.

Persistent memory and multi-agent communication enlarge the governance surface. Stored or repeated assertions do not become trusted by repetition, and messages do not create delegation merely because agents can communicate. Stateful correlation is therefore evaluator state, not a twenty-seventh FIP primitive.

The architecture is a semantic analogue of zero-trust reasoning: semantic intelligibility does not imply operational authority. This is an analogy, not NIST terminology and not a replacement for network zero trust.

The protected experiments observed zero unauthorized operational effects under their bounded conditions. This does not imply a zero probability of failure, universal security,

formal verification, FIP completeness or minimality, production certification, universal domain independence, or prompt-injection prevention.

Governance friction is also a substantive deployment question. In Q12-B3, 256 of 541 decisions (47.3%) were INCOMPLETE. INCOMPLETE is not equivalent to DENIED: it means required trusted governance meaning was absent and the fail-closed evaluator refused to infer permission. Because Q12-B3 was an adversarial workload, this rate should not be interpreted as an expected production rate. Future benign-workload studies should measure INCOMPLETE/ESCALATE frequency together with latency, false denial, successful task completion, and human-review burden.

The generative adversary was constrained and did not test trusted-computing-base compromise. GOCP also assumes legitimate configuration; a human-approved incorrect mapping can still cause incorrect governance. Future work should examine independent approval, profile-evolution attacks, richer feedback, longer-horizon multi-agent collusion, realistic physical/tool boundaries, formal verification of selected invariants, and governance-friction metrics.

X. RELATED WORK

Reference monitors and capability security. Classical protection work established complete mediation and capability-based bounded authority [1], [2]. The SMF/gateway inherits this lineage but evaluates richer interaction semantics and authority bindings at an autonomous effect boundary.

Trust management, RBAC, ABAC, XACML, and deontic policy. PolicyMaker and related trust-management work separated credentials, policy, and trust decisions [3]. RBAC [4], ABAC [5], and XACML [6] provide mature authorization models. AgenticRei [15] extends runtime governance with deontic permissions, prohibitions, obligations, dispensations, conflict resolution, and ontological reasoning. FIP/GOCP does not seek to replace these policy formalisms. Its distinct upstream question is which assertions from an extensible semantic environment are permitted to become governance-bearing inputs to such a decision space, and how the resulting authority remains bound under composition.

ODRL and provenance. ODRL represents permissions, prohibitions, duties, actions, assets, parties, and constraints [7]. W3C PROV provides interoperable provenance semantics [8]. FIP reuses concepts from these established families but distinguishes represented provenance from trusted provenance and treats normative concepts as part of a consequential interaction envelope.

Agent communication languages. KQML [9] and FIPA ACL [10] establish structured communicative acts. FIP is not a transport or dialogue language; its distinction is that a communicative act does not itself establish operational authorization.

Zero trust. NIST SP 800-207 rejects implicit trust based on network location [11]. The proposed semantic analogy rejects implicit operational authority based merely on semantic intelligibility.

Runtime assurance and AI control. Simplex-style runtime assurance separates advanced autonomy from an independent safety mechanism; recent Synergistic Simplex work extends that pattern [12]. AI Control studies protocols intended to

prevent harmful outcomes even when a powerful model may intentionally subvert safeguards [13]. The present work applies an independent runtime-control pattern specifically to semantic authorization and operational consequence.

Recent agentic-AI governance. Governance by Construction places policy interventions at multiple points in an agent pipeline [14]. AgenticRei develops deontic runtime policies for permissions, prohibitions, obligations, dispensations, and conflict resolution [15]. SARC compiles governance constraints into runtime enforcement sites and extends them to multi-agent workflows [16]. Aegis treats model outputs as action proposals mediated by trusted runtime policy and provenance [17]. TokenWall acts as a semantic firewall over persistent-agent token flows [18]. CAVA canonicalizes heterogeneous runtime actions for approval binding and attestation [19]. MI9 proposes semantic telemetry, continuous authorization, conformance, drift detection, and containment [20].

Positioning. The term semantic firewall and the individual concepts of runtime mediation, deontic policy, provenance, complete mediation, and structured agent acts all have prior art. The proposed contribution is compositional rather than a claim to invent those components: an explicit governed descriptive-to-operational projection (GOCP); a compact consequential-interaction vocabulary (FIP); stateful evaluation of bound authority including authority recomposition; and enforcement at the actual effect boundary. The contribution is therefore not another policy engine, but an explicit governed boundary controlling how open semantics become operational authority, followed by bound authorization and effect-boundary enforcement.

Table II gives a conservative comparison of representative contemporaneous systems. It summarizes claims explicit in the cited papers; omission of a feature is not a claim that the system cannot implement it.

TABLE II. Representative runtime-governance systems and positioning.

System / runtime-governance focus	Position relative to FIP/GOCP
CUGA [14] - Policy-as-code at five pipeline checkpoints, including tool approval.	Typed pipeline governance; FIP/GOCP isolates semantic admission and bound interaction authority.
AgenticRei [15] - External OWL/Rei engine for deontic policy, conflict resolution, and ontology reasoning.	Reasoning after predicates enter the decision space; GOCP governs which open semantics may become governance-bearing inputs.
SARC [16] - Pre-action, action-time, post-action, and escalation sites; multi-agent constraint propagation.	First-class runtime constraints and authority intersection; FIP adds explicit 26/24 interaction semantics and GOCP admission.
Aegis [17] - Trusted action-proposal mediation with server-side provenance and fail-closed/quorum settlement.	Closest overlap at the effect boundary; this work also makes descriptive-to-operational semantic admission an explicit governed artifact.
TokenWall [18] - Semantic auditing of token flows before privileged runtime sinks.	Flow-risk classification rather than a governance-bearing interaction-role vocabulary and authority-binding model.
CAVA [19] - Canonical action identity, approval binding, receipts, portability, and attestation.	Canonicalizes heterogeneous actions; FIP/GOCP focuses on interaction semantics plus governed contextual authority.
MI9 [20] - Semantic telemetry, continuous authorization, FSM conformance, drift detection, and containment.	Integrated runtime governance; this work emphasizes semantic non-authority, governed projection, and recomposition attacks.
This work - GOCP → FIP → stateful SMF → enforced gateway.	Explicit semantic admission boundary; bound authority; qualified symbolic + Luna + Sol adversarial evaluation; open artifact [25].

Artifact Availability—The FIP 0.1 specification, GOCP schema, Semantic Micro-Firewall reference implementation, 47 frozen conformance vectors, adversarial experiment definitions, Q12-C/Q12-C2 evidence, contribution guidance, and reproduction instructions are publicly available in the Interbilla FIP repository [25]: [Interbilla FIP repository](#). No external independent implementation is claimed at the time of this manuscript.

XI. CONCLUSION

As autonomous AI systems invoke tools, modify persistent state, and coordinate across organizations, governance must shift its focus. The critical question is no longer whether a model understands an instruction, but whether a specific interaction is authorized to produce a specific effect. Our architecture addresses this by separating descriptive semantics from operational authority, enforcing a central rule: open description, closed operational interpretation.

Within this framework, FIP 0.1 serves as a 26-primitive, 24-relation reference candidate rather than a rigid universal standard. To prevent arbitrary data from seizing authority, the GOCP blocks descriptive semantics from carrying governance weight without an explicit mapping. Meanwhile, the SMF evaluates authority as a bound relation instead of an ambient property, and the resource gateway anchors that decision to the actual consequence.

We stress-tested this architecture against both symbolic and generative adversaries. Even when pitted against a stronger GPT-5.6 Sol replication—which demonstrated higher adversarial effectiveness than Luna—the system allowed zero unauthorized operational effects across all evaluated domains, attack classes, and budgets. While this empirical evidence motivates independent testing, it does not close the security question. Our stance on FIP 0.1 is explicit: we are not asking the community to accept it; we are asking the community to test it. The public artifact [25] includes full reproducibility material. The most valuable next step for this research is for the community to find a consequential interaction that breaks our proposed authority boundary, requires a new primitive, or exposes a redundant one.

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Status note: FIP 0.1 is a Reference Candidate proposed for community testing and improvement. It is not a published standard or production certification.